

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	P. Sai Charan Tej	Reg. No.:	192465048
Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

1. An advertisement platform ranks ads based on relevance and bid value. Analyze how Quick Sort can be applied for efficient ranking. Identify issues such as pivot selection and data distribution. Apply divide-and-conquer techniques, analyze the performance, and propose optimizations with justification.

Code of Conduct

I P. Sai Charan Tej(192465048)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Algorithm: Online Advertisement Ranking using Quick Sort

1. **Start.**
2. Define functions `compute_score()`, `median_of_three()`, `partition()`, `quick_sort()`, and `insertion_sort()`.
3. Read the number of advertisements.
4. Read the **advertisement name, relevance score, and bid value.**
5. Calculate the combined score using:
 - Relevance weight = **0.6**
 - Bid weight = **0.4**
6. Store the calculated score along with the advertisement details.
7. Select the **median-of-three pivot** for better pivot selection.
8. Partition the advertisements based on the score in **descending order.**
9. Recursively apply Quick Sort to the smaller partition.
10. Use **insertion sort for small partitions** to improve performance.
11. Repeat the sorting process until all advertisements are ranked.
12. Display the **rank, advertisement name, score, relevance, and bid value.**
13. Record and analyze the execution performance if required.
14. **Stop.**

Time Complexity: Average – $O(n \log n)$, Worst – $O(n^2)$

Space Complexity: $O(\log n)$ due to optimized recursion.

Output:-



SIMATS Online Programming Lab

DAA PROGRAMMING

 POTHURAJU SAI CHARAN TEJ
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QUESTIONS

Q1

Online Advertisement Ranking (Quick Sort) An advertisement platform ranks ads based on relevance and bid value
100 marks

0/1 answered

Q1

Online Advertisement Ranking (Quick Sort)

An advertisement platform ranks ads based on relevance and bid value

100 marks

Analyze how Quick Sort can be applied for efficient ranking. Identify issues such as pivot selection and data distribution. Apply divide and conquer techniques. Analyze performance and propose optimizations with justification.

Your Code

```
1 def median_of_three(arr, low, high):
2     mid = (low + high) // 2
3     if arr[low][0] > arr[mid][0]:
4         arr[low], arr[mid] = arr[mid], arr[low]
5     if arr[low][0] > arr[high][0]:
6         arr[low], arr[high] = arr[high], arr[low]
7     if arr[mid][0] > arr[high][0]:
8         arr[mid], arr[high] = arr[high], arr[mid]
9     arr[mid], arr[high - 1] = arr[high - 1], arr[mid]
10    return arr[high - 1][0]
11
12
13 def partition(arr, low, high):
14     if high - low > 2:
15         pivot = median_of_three(arr, low, high)
```

Input

```
5
Ad1 88 50
Ad2 98 78
Ad3 68 188
```

Output

```
Rank | Ad Name | Score |
Relevance | Bid Value
1 | Ad2 | 92.00 | 90.0 | 70.0
2 | Ad5 | 77.00 | 75.0 | 88.0
3 | Ad3 | 76.00 | 60.0 |
400.0
```